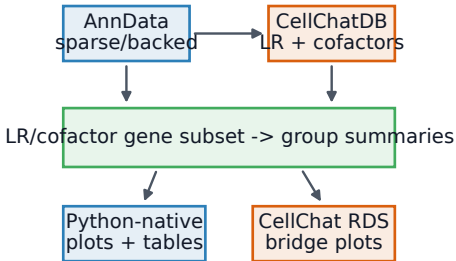


AnnData-first CCC workflow



Shared LR table and grouped expression keep results comparable while avoiding CellChat's cell-level dense path.